

CHAPTER 1

INTRODUCTION

IoT-based smart waste management uses sensors to detect bin fill level and send alerts in real-time. It helps collect waste on time, prevents overflow, and makes the process more efficient. This system improves cleanliness and reduces manual effort.

The rapid growth of urbanization has significantly increased the amount of solid waste generated in cities, creating challenges for efficient waste collection and disposal. Traditional waste management methods often rely on fixed collection schedules that do not consider the real-time fill level of waste bins, leading to overflowing bins, unhygienic surroundings, and unnecessary fuel consumption. To overcome these limitations, Internet of Things (IoT) technology has emerged as a smart and scalable solution for automating the waste monitoring and collection process [1].

IoT-based systems incorporate a network of sensors, microcontrollers, and wireless communication modules to continuously monitor waste levels and transmit data to a central platform. These systems enable municipal authorities to make informed decisions, optimize collection routes, and reduce operational costs. By using ultrasonic sensors, GSM/GPRS modules, and cloud platforms, smart waste bins can send alerts when they reach a certain threshold level, ensuring timely disposal and improved cleanliness in public areas [2].

A smart waste management system not only enhances efficiency but also supports environmental sustainability. With real-time monitoring, the frequency of collection trips can be minimized, thereby reducing carbon emissions and fuel consumption. IoT-enabled dashboards allow administrators to track bin fill-levels, analyze waste patterns, and plan resource allocation. This improves the responsiveness of waste collection teams and reduces the chances of overfilled bins or delayed garbage pickup [3].

Furthermore, the integration of GPS, ESP32/ESP8266 microcontrollers, and communication technologies supports remote tracking and automated notifications. Such IoT-based waste monitoring systems are suitable for smart cities, residential societies, educational institutions, and commercial zones. They provide a reliable and low-cost solution for managing solid waste more intelligently by combining sensing technology, wireless communication, and cloud-based data analytics [4].

To ensure long-term operation, IoT nodes require low-power components, efficient system design, and minimal maintenance. Optimized hardware configuration and energy-efficient communication protocols play a crucial role in enhancing system performance and durability. These design considerations make IoT-based waste management systems a practical and scalable approach to maintaining urban hygiene and supporting sustainable development goals [5].

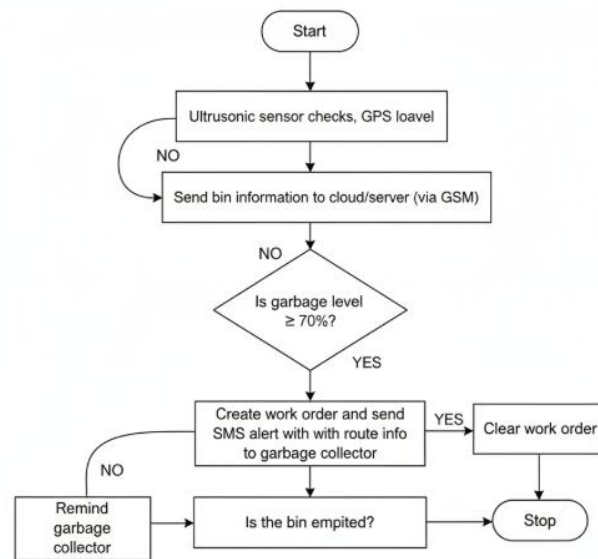


Fig. 1.1 Block diagram of the proposed system.

The flowchart explains how a smart waste management system operates. First, the system starts and continuously monitors the dustbin using an ultrasonic sensor to measure the garbage level, while GPS tracks its location. The measured data is sent to the cloud or server through GSM communication. If the garbage level is below 70%, the system continues monitoring normally. However, if the garbage level becomes 70% or above, the system automatically generates a work order and sends an alert message along with the route information to the garbage collector. After informing, the system waits and checks whether the garbage has been cleared. If the bin remains full, it sends reminders to the collector until the bin is emptied. Once the garbage is collected and the bin is cleared, the work order is removed and the process stops.

CHAPTER 2

LITERATURE REVIEW

Fathima et al. [6], IoT sensors integrated with microcontrollers can effectively monitor garbage levels in bins and send real-time alerts to municipal authorities. Their study emphasizes that smart waste monitoring helps reduce manual inspection, prevents overflow, and supports smart city automation.

Vamsi et al. [7] proposed an IoT-based smart garbage monitoring and disposal support system that uses ultrasonic sensors and GSM modules for automatic communication. Their design enables authorities to know when a bin is nearly full, ensuring timely waste collection. The system significantly reduces human effort and enhances the efficiency of solid-waste management operations.

Sosunova and Porras [8] highlights that IoT-enabled waste management solutions are transforming smart cities by improving data collection, route optimization, and environmental monitoring. Their study found that integrating cloud platforms with sensor networks allows large-scale monitoring and supports more sustainable waste handling practices.

This block diagram shows how the smart bin node works: the bin level is first sensed and converted into a clean signal by the amplifying or conditioning circuit, then the ESP32 microcontroller reads this signal and uses its WiFi or cloud module to send the processed bin status to the online platform for monitoring and alerts.

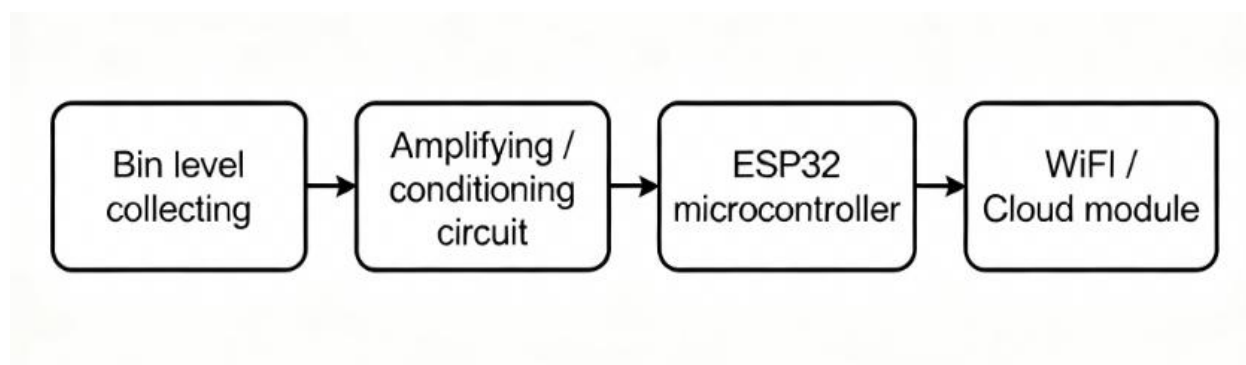


Fig. 2.1 Schematic diagram.

Rahman et al. [9] introduced an intelligent garbage management model that not only monitors waste levels but also focuses on fairness in waste collection across different city zones. Their IoT-based architecture ensures balanced distribution of collection tasks and avoids biased or delayed garbage pickup. This contributes to improved hygiene and smart city governance.

Ghahramani et al. [10] proposed a route recommendation system using IoT data to optimize waste collection paths for garbage trucks. Their approach uses real-time bin-level data and geographical information to minimize travel distance, reduce fuel consumption, and enhance collection efficiency. This demonstrates that IoT plays a crucial role in intelligent decision-making for modern waste management systems.

CHAPTER 3

PROBLEM DEFINITION

Rapid urbanization has increased the amount of waste generated in cities, but traditional waste collection methods have not adapted to this growth. Garbage bins are still emptied on fixed schedules, regardless of how full they are. As a result, bins often overflow when trucks arrive late or remain half-empty when collected too early. This leads to unhygienic conditions, foul odor, pollution, and health risks for the public. Inefficient truck routes also cause fuel wastage, increased operational costs, and unnecessary traffic congestion.

A major issue is the lack of real-time data, which prevents authorities from knowing which bins need immediate attention. Some areas receive timely collection while others face delays, creating inequitable waste management services. Manual monitoring is time-consuming, inaccurate, and labor-intensive.

Therefore, there is a need for a smart system that uses IoT technology to monitor bin levels, send alerts, and optimize collection routes, ensuring cleaner, more efficient, and sustainable waste management.

CHAPTER 4

METHODOLOGY

The IoT-based smart waste management system uses sensors, a microcontroller, and wireless communication to monitor bins in real time. Ultrasonic sensors measure the waste level, while IR sensors detect when someone approaches the bin. The ESP32 microcontroller processes these sensor readings, controls the lid mechanism through a servo motor for touch-free operation, and sends data via its built-in Wi-Fi.

The cloud platform receives information such as fill level, lid activity, and usage frequency. When the bin reaches a set limit, alerts are generated to inform waste collection teams. This smart system helps prevent overflow, improves cleanliness, and allows municipal authorities to plan efficient collection routes and schedules based on real-time data [11].

4.1HARDWARE REQUIREMENT:

The system hardware consists of the following components:

a) ULTRASONIC SENSOR (HC-SR04):

The HC-SR04 ultrasonic sensor is used to measure the garbage level in the dustbin by calculating the distance between the sensor and the waste surface. It sends ultrasonic waves from the TRIG pin and receives the reflected wave on the ECHO pin. The ESP32 calculates the distance and converts it into a fill percentage for monitoring.



Fig. 5.1 HC-SR04 Ultrasonic Sensor

CONNECTION OF DEVICE DEVICE:

HC-SR04 Pin	Connections
VCC	5V power supply
GND	GND on ESP32
TRIG	GPIO 13
ECHO	GPIO 14

b) ESP32 MICROCONTROLLER:

The ESP32 is a powerful microcontroller featuring a dual-core processor, built-in Wi-Fi, and Bluetooth connectivity. In this project, it is used to read ultrasonic sensor data, display the waste level on the LCD, and send alerts to the cloud using Blynk. Its multiple GPIO pins and high-speed processing make it suitable for real-time IoT waste monitoring applications.

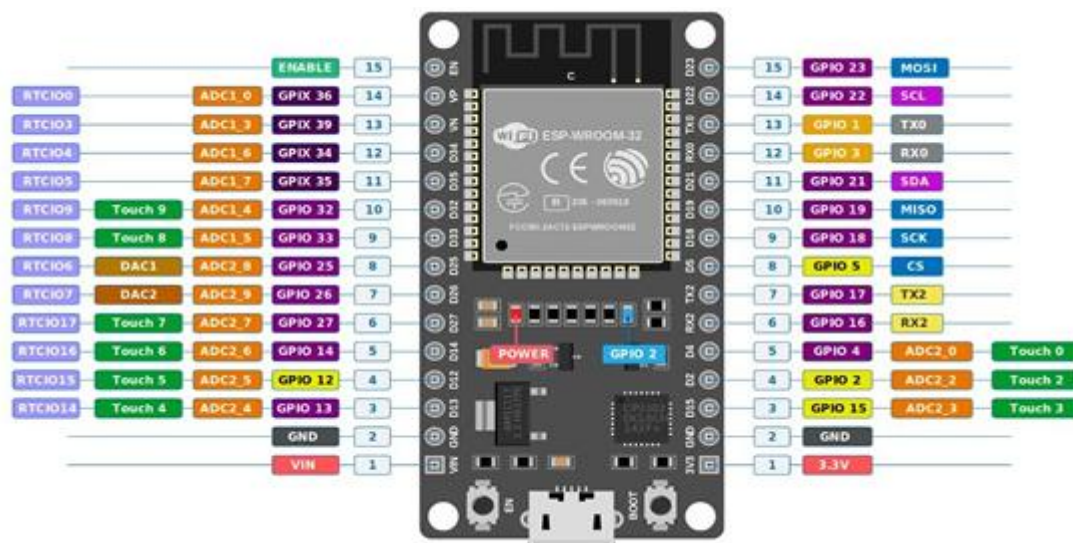


Fig. 5.2 ESP32 MICROCONTROLLER

c) NEO-6M GPS MODULE:

The NEO-6M GPS module provides real-time location coordinates (latitude and longitude) of the smart dustbin. This information is sent to Blynk Map or displayed when alerts are generated. It communicates with ESP32 using UART (TX/RX).

CONNECTION OF HARDWARE DEVICE:

VCC	3.3V pin of ESP32 (3V3)
GND	GND of ESP32
TX	RX2 pin of ESP32 (GPIO 16)
RX	TX2 pin of ESP32 (GPIO 17)



Fig. 5.3 NEO-6M GPS MODULE.

d) SIM900 GSM MODULE:

The SIM900 GSM module is used to send SMS alerts when the dustbin reaches critical fill levels. It communicates with the ESP32 via UART and operates on a 5V power supply. The module enables remote monitoring even without Wi-Fi.

CONNECTIONS OF HARDWARE SYSTEM:

VCC (VBAT)	5V external supply
GND	GND of ESP32
TXD	GPIO 16 (RX2 of ESP32)
RXD	GPIO 17 (TX2 of ESP32)



Fig. 5.4 SIM900 GSM MODULE.

e) I2C 16×2 LCD DISPLAY:

The LCD display (16×2) with I2C module is used to show the real-time distance, fill percentage, and system status. The I2C interface reduces wiring and uses only two pins of the ESP32 (SDA & SCL).

CONNECTION OF I2C 16x2 DISPLAY:

VCC	ESP32 5V (VIN)
GND	ESP32 GND
SDA	ESP32 GPIO 21
SCL	ESP32 GPIO 22



Fig. 5.5 I2C 16×2 LCD DISPLAY.

e) POWER SUPPLY (5V):

A regulated 5V power supply is used to power modules like GSM, GPS, LCD, and ultrasonic sensor. The ESP32 can also be powered through 5V input. A stable power supply ensures reliable operation of all components.



Fig. 5.6 POWER SUPPLY (5V).

f) USB-C Interface for ESP32 Power and Data Transfer:

A USB-C interface on an ESP32 development board is a modern USB Type-C connector combined with a USB-to-serial chip that allows a single cable to provide both 5 V power and data communication with a computer. When you plug the USB-C cable in, the board's voltage regulator converts the 5 V from USB to 3.3 V for the ESP32, so you usually do not need a separate power supply during development. At the same time, the USB-to-UART bridge exposes a virtual COM port on the PC, letting you upload code, view Serial Monitor output, and debug the ESP32 directly through the USB-C connection.



Fig. 5.7 USB-C Interface for ESP32 Power and Data Transfer.

•4.1.1 Schematic diagram of ESP32-based IoT smart waste bin

The smart waste bin node is built around an ESP32 development board that acts as the main controller and provides in-built Wi-Fi connectivity for future cloud or mobile integration. The ESP32 is powered from a regulated 3.3 V line derived from a common 5 V DC supply, while its GPIO pins interface with all peripheral modules and share a common ground reference.

For level measurement inside the bin, an HC-SR04 ultrasonic sensor is mounted at the top opening and connected to the ESP32 using two digital pins: one configured as TRIG (for transmitting the ultrasonic pulse) and another as ECHO (for receiving the reflected signal). The sensor is supplied from the 5 V rail with its ground tied to the system ground, and the measured echo time is processed by the ESP32 to calculate the fill level of the bin.

To provide location information, a Neo-6M GPS receiver module is interfaced to the ESP32 through a UART interface, typically using one GPIO as RX and another as TX. The Neo-6M is powered either from the 3.3 V output of the ESP32 (or a dedicated 3.3 V regulator if required), and continuously streams NMEA data which the ESP32 parses to obtain latitude and longitude of the bin.

For remote alerts, a SIM900A GSM module is connected to a second UART port of the ESP32 so that AT commands can be used to send SMS messages to the concerned authority or garbage collector when the bin is full or in fault condition. The SIM900A draws its supply from the 5 V line (often via a separate high-current regulator) with common ground, while logic-level connections are made to the ESP32's TX and RX pins.

A 16×2 alphanumeric LCD with an I2C backpack is used for local status indication such as current bin level, network status, and error messages. The I2C lines, SDA and SCL, are connected to the ESP32's I2C-capable GPIO pins, and the LCD module is powered from the 5 V rail, with the I2C backpack handling level compatibility while sharing the same ground.

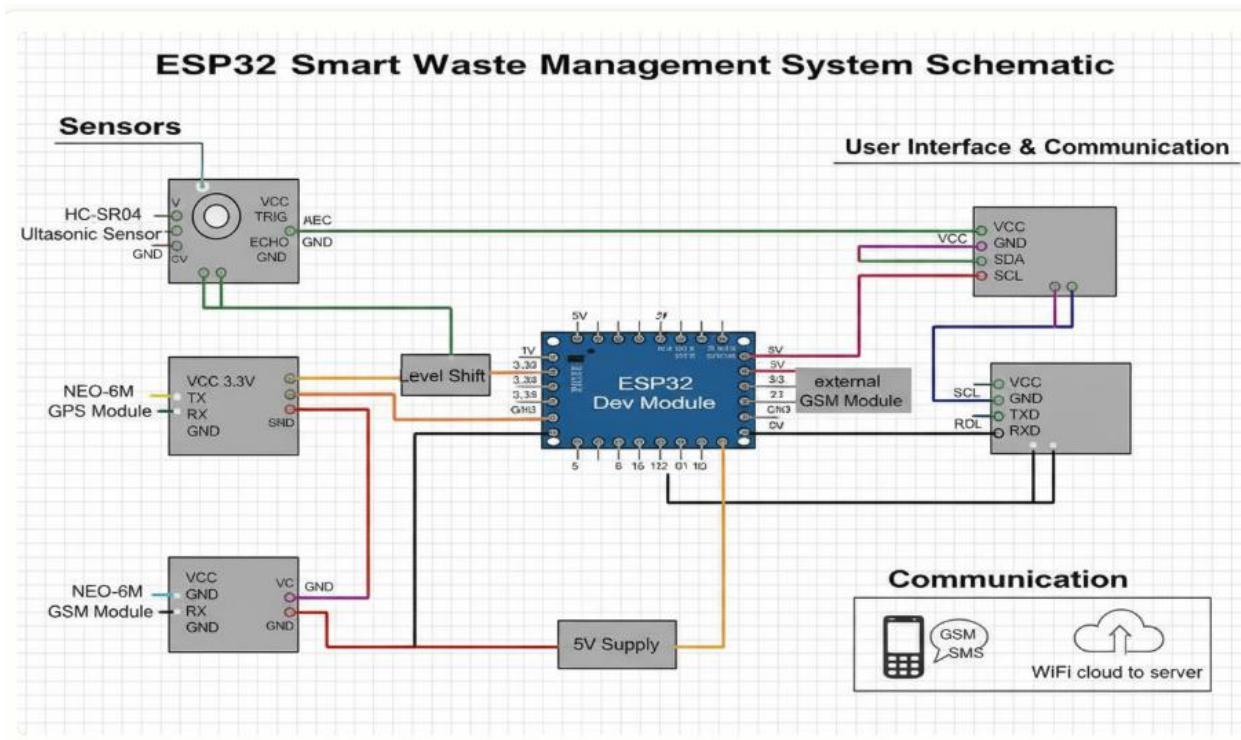


Fig. 5.8 Schematic representation of ESP32-based smart waste bin node.

This figure shows the ESP32 placed at the center with clearly labeled connections to each peripheral and the power rails. The HC-SR04 is linked to two digital GPIO pins on the ESP32 for TRIG and ECHO, the Neo-6M GPS and SIM900A GSM modules are each connected via separate UART pairs, and the I2C 16×2 LCD is tied to the ESP32 through its SDA and SCL pins. All modules receive power from a common 5 V supply, from which a 3.3 V regulator feeds the ESP32, and a shared ground line ensures proper reference for every signal path in the node.

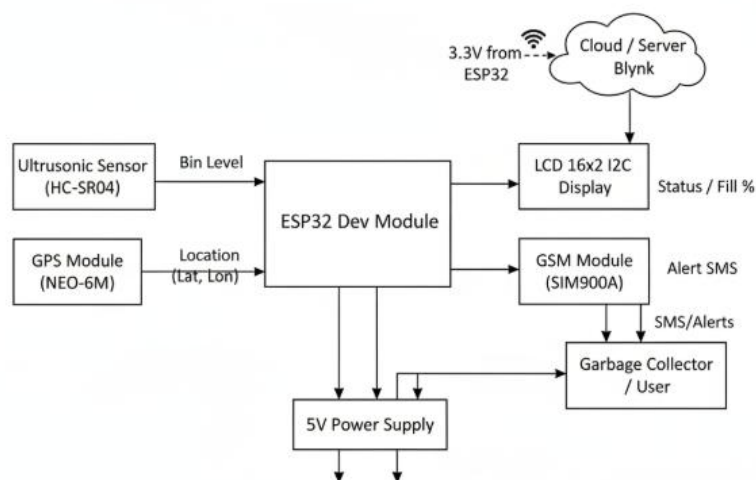


Fig. 5.8 Schematic representation of hardware implementation of the system.

The hardware block diagram abstracts the node into functional blocks: Power Supply, ESP32 Controller, Ultrasonic Sensor, GPS Module, GSM Module, and LCD Display. Arrows from the Power Supply block distribute 5 V (and the derived 3.3 V) to each module, while signal arrows indicate that the Ultrasonic Sensor, GPS, GSM, and LCD all communicate with the ESP32, which in turn can forward data to a Cloud/Server/Blynk block. From this cloud block, another arrow leads to a “Garbage Collector/User” block, representing how processed bin status and location information are finally delivered as notifications or monitoring data.

4.2 SOFTWARE REQUIREMENT:

For the software requirements of this project, the Arduino IDE is used as the main development environment to write, compile and upload the embedded C/C++ program to the ESP32 microcontroller. The ESP32 board support package must be installed in the IDE so that the correct compiler, uploader and pin definitions are available for the board. In addition, several libraries are required, such as the WiFi library for network connectivity, a library to interface the ultrasonic sensor, a GPS library for reading NEO-6M location data, a GSM library for controlling the SIM900A module using AT commands, and an I2C LCD library to drive the 16×2 display with minimal code. The computer must also have the appropriate USB-to-serial driver (for example CP2102 or CH340, depending on the ESP32 board) so that the board appears as a serial COM port during programming and debugging. If cloud or IoT services are used, such as Blynk or an HTTP API, the required accounts, authentication tokens and network settings are configured in the program to enable remote monitoring and alert notifications.

a) ARDUINO IDE:

The Arduino Integrated Development Environment (IDE) is used to develop and upload the embedded C/C++ program for the ESP32 microcontroller. It provides a code editor, compiler and uploader in a single tool, and also offers the Serial Monitor window to view sensor readings, GPS data and debug messages while the system is running. The same sketch that is tested in Wokwi can later be compiled in the Arduino IDE and programmed into the real ESP32 board.

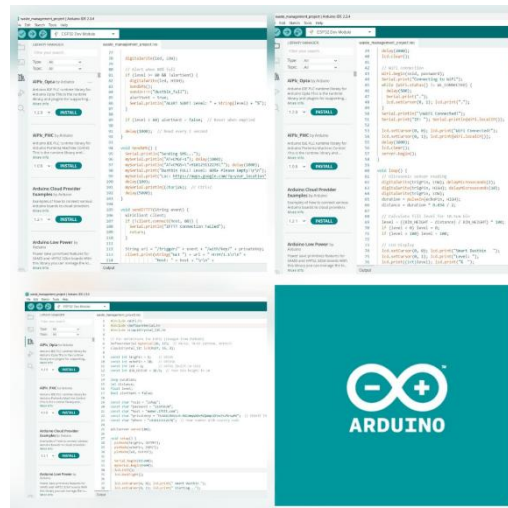


Fig.5.9 Arduino IDE code implementation for the proposed system

b) ESP32 Board Package and Libraries:

The ESP32 board support package is installed in the Arduino IDE using the Boards Manager so that the IDE recognizes the ESP32 Dev Module, its GPIO pins and upload settings. This package also provides the core WiFi and hardware libraries needed by the sketch. The WiFi library can be used for future cloud connectivity, while the Wire (I2C) library is used to communicate with the 16×2 LCD display and any other I2C devices in the project.

c) External Libraries for Modules:

Several external libraries are included to interface the different modules easily. A GPS library such as TinyGPS++ is used to read the NEO-6M GPS serial data and convert it into human-readable latitude and longitude. A GSM library for the SIM900A module simplifies sending AT commands so that SMS alerts can be sent with simple function calls instead of raw serial strings. An I2C LCD library (for example LiquidCrystal_I2C) provides high-level functions like `lcd.print()` to display text on the 16×2 LCD using only the SDA and SCL lines of the ESP32.

d) Wokwi Online Stimulator:

Wokwi is an online simulation environment used in this project to test the ESP32 smart waste bin when the physical hardware was not fully working. The same Arduino sketch is uploaded to a virtual ESP32 in Wokwi, where virtual ultrasonic sensors, GPS modules, GSM modules and LCDs can be connected. This allows checking the logic of fill-level calculation, GPS handling and SMS alert generation before deploying the code on the real prototype.

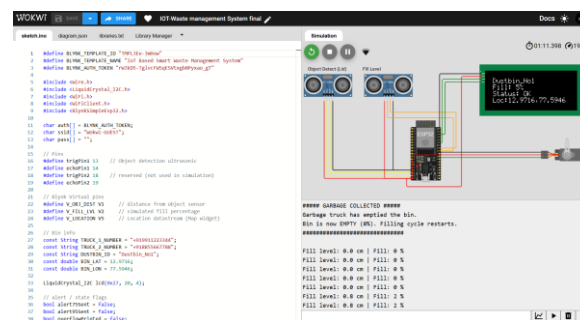


Fig. 5.9 Wokwi simulation of the ESP32-based smart waste bin node.

CHAPTER 5

RESULT AND DISCUSSIONS

The proposed IoT-based smart waste management system was implemented using an ESP32 microcontroller, HC-SR04 ultrasonic sensor, NEO-6M GPS module, SIM900A GSM module and I2C 16×2 LCD. The hardware prototype was assembled and tested on a dustbin, and the same circuit was also created in the Wokwi ESP32 simulator for reliable functional verification. The main objective of this chapter is to present the behaviour of the system under different bin-fill conditions and to discuss how accurately the node detects the level, updates the display and generates alert messages.

5.1 Experimental hardware setup

The experimental setup of the ESP32-based smart waste bin node is shown in Fig. 5.x. The ESP32 development board is interfaced with the ultrasonic level sensor fixed at the top of the bin, while the GPS, GSM module and I2C LCD are mounted on a breadboard and powered from a regulated 5 V supply. The ultrasonic sensor measures the distance from the sensor to the garbage surface; this value is read by the ESP32 and converted into a percentage of the bin height. The LCD displays the dustbin ID, fill percentage, status and GPS coordinates, and the GSM module is configured to send SMS alerts when the bin crosses predefined thresholds. During testing some hardware issues (mainly power instability and GSM network problems) limited continuous operation, therefore the detailed functional behaviour was evaluated using the Wokwi simulation of the same circuit.

5.2 Wokwi simulation prototype

To validate the logic independent of hardware issues, the complete circuit was recreated in the Wokwi ESP32 simulator. Fig. 5.y shows the Wokwi simulation prototype of the system at different fill levels. The virtual circuit contains the ESP32, two ultrasonic sensors (lid detection and fill level) and the I2C LCD. Four screenshots are captured for four stages: low, moderate, almost full and critical. When the bin is almost empty, the LCD shows a low fill percentage (around 10%) with the status “OK”. As the simulated level increases to around 60–70%, the fill percentage increases and the status changes to “Moderate”. When the level reaches about 80–90%, the display shows “Almost”, indicating that the bin is close to full. Finally, when the level exceeds about 95%, the status becomes “Critical”, clearly indicating that immediate collection is required. In all screens the dustbin ID and GPS location remain visible, confirming that the program correctly updates the user interface according to the measured fill level.

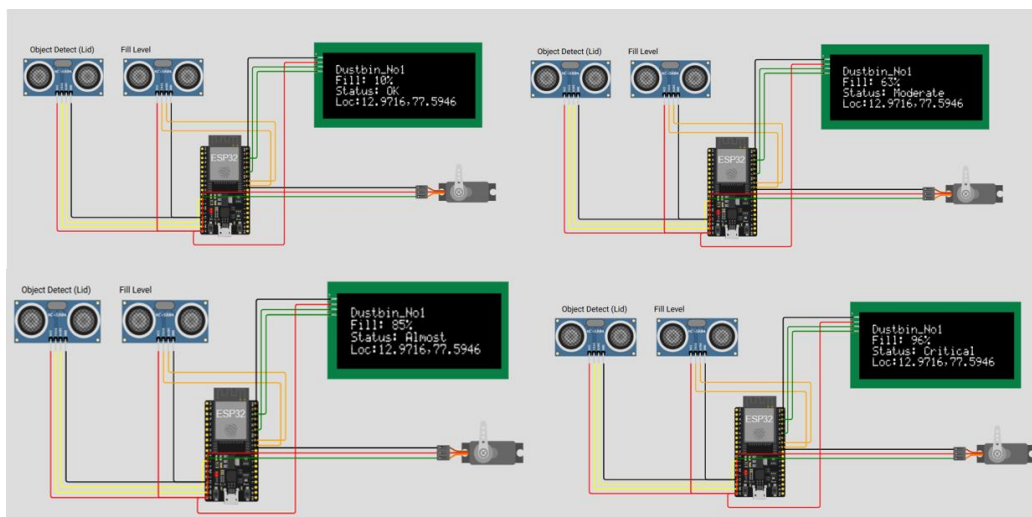


Fig. 5.y Wokwi simulation prototype of smart waste bin at different fill levels.

5.3 LCD output at various levels

For clarity, the LCD outputs at each stage are also presented separately as Fig. 5.y.1 to Fig. 5.y.4. At low level the display shows “Fill: 10% Status: OK”, at moderate level it shows around “Fill: 60% Status: Moderate”, at almost full condition it shows “Fill: ~85% Status: Almost”, and at critical level it shows “Fill: ~96% Status: Critical”. Along with these messages, the dustbin ID and latitude–longitude pair are printed on the last line. These screenshots demonstrate that the ESP32 continuously calculates the fill percentage from the ultrasonic readings and changes the status message in real time, giving a clear visual indication of bin condition even in the simulated environment.

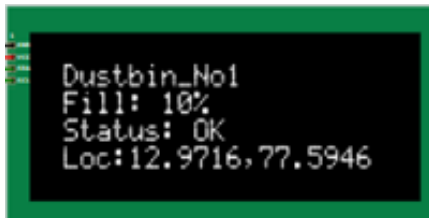


Fig. 5.a – LCD display at low level (OK)

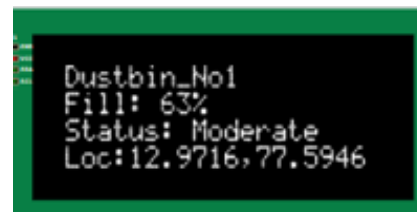


Fig. 5.b – LCD display at moderate level

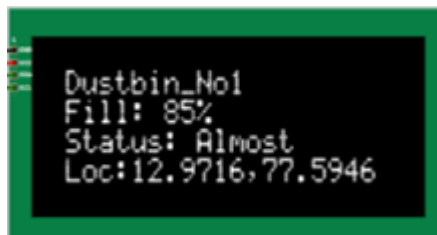


Fig. 5.c – LCD display at almost full level

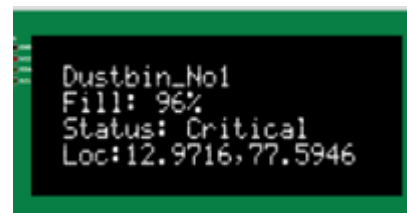


Fig. 5.d – LCD display at critical level

Fig. 5.a.1–5.b.4 LCD display outputs for OK, Moderate, Almost and Critical states.

5.4 Serial Monitor output and alert behaviour

The detailed behaviour of the algorithm is observed from the Serial Monitor output, shown in Fig. 5.z. The log lists the calculated fill height in centimetres and the corresponding fill percentage for each measurement cycle, proving that the conversion from distance to percentage is working correctly. When the percentage first reaches 75%, the program generates a “75% LEVEL REACHED” warning message and prints the SMS text that would be sent to Truck 1, including the bin fill percentage and a Google Maps link with the GPS coordinates. As the level continues to rise and crosses 95%, a final critical alert message is produced, indicating that all available garbage collectors are notified and again providing the location link. After a simulated collection event, the Serial Monitor shows the message “GARBAGE COLLECTED” and resets the fill level to 0%, after which a new filling cycle starts. These results confirm that the threshold comparison, alert generation and reset logic of the system are functioning as intended.

```

Fill level: 30.0 cm | Fill: 75 %
--- ALERT: 75% LEVEL REACHED ---
Truck 1 is notified to collect this bin.
Fill: 75 %
Maps: https://maps.google.com/?q=12.971600,77.594600
-----

Fill level: 30.0 cm | Fill: 75 %
Fill level: 30.0 cm | Fill: 75 %
Fill level: 30.0 cm | Fill: 75 %
Fill level: 30.0 cm | Fill: 75 %
Fill level: 30.4 cm | Fill: 76 %
Fill level: 30.4 cm | Fill: 76 %
Fill level: 30.4 cm | Fill: 76 %
Fill level: 30.4 cm | Fill: 76 %
Fill level: 32.0 cm | Fill: 80 %
Fill level: 32.0 cm | Fill: 80 %
Fill level: 32.0 cm | Fill: 80 %
Fill level: 32.0 cm | Fill: 80 %
Fill level: 32.0 cm | Fill: 80 %
Fill level: 33.2 cm | Fill: 83 %
Fill level: 33.2 cm | Fill: 83 %
Fill level: 33.2 cm | Fill: 83 %
Fill level: 35.6 cm | Fill: 89 %
Fill level: 35.6 cm | Fill: 89 %

=== FINAL ALERT: 95% LEVEL ===
All available garbage collectors (Truck 1 & 2) notified.
Fill: 98 %
Maps: https://maps.google.com/?q=12.971600,77.594600
=====

Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %
Fill level: 39.2 cm | Fill: 98 %
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Fill level: 39.2 cm | Fill: 98 %

##### GARBAGE COLLECTED #####
Garbage truck has emptied the bin.
Bin is now EMPTY (0%). Filling cycle restarts.
#####

```

Fig. 5. Serial Monitor output showing real-time fill level and alert messages in simulation.

5.5 Discussion

From the above observations it can be seen that the proposed node is able to continuously monitor the fill level of the dustbin and correctly distinguish between normal, warning and critical conditions. The LCD interface provides an intuitive indication of bin status for users near the bin, while the GSM alert text prepared in the code ensures that the responsible garbage-collection staff receive timely information about bins that require attention, together with their exact GPS location. Although full validation on physical hardware was limited by practical constraints, the Wokwi-based experiments demonstrate that the control logic, communication format and user interface of the ESP32 smart waste bin system work reliably and can be directly deployed once stable hardware and power supply are available.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

The proposed IoT-based smart waste management system successfully demonstrates how ESP32, ultrasonic sensing, GPS tracking, GSM communication and an LCD interface can be combined to monitor dustbin status in real time and automate notifications. By continuously measuring the fill level with the HC-SR04 sensor and associating it with accurate location data from the Neo-6M GPS module, the system can identify which bins are nearly full and where they are placed. The SIM900A GSM module enables automatic SMS alerts to be sent to the concerned garbage collector only when threshold levels are crossed, reducing unnecessary collection trips and helping to optimize fuel and manpower. The use of an I2C 16×2 LCD provides local feedback to users near the bin, while the ESP32 controller and the developed software ensure reliable decision-making based on sensor readings. Overall, the project meets its objective of making waste collection smarter, more efficient and closer to real-time, and the same architecture can be extended in future to support cloud dashboards or larger networks of smart bins in a smart-city environment.

6.2 Future work

In future, this project can be improved in several simple ways. The system can be connected to a cloud dashboard so that many bins in a city are monitored on a single map and collection routes are planned automatically. The SMS alert method can be extended with mobile-app or web notifications, making it easier for staff to track and confirm each collection. More sensors such as gas or temperature sensors can be added to detect bad smell or fire inside the bin for safety. Low-power modes and a small solar panel can also be used so that the smart bin works reliably outdoors for long periods without frequent charging.

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